

# Daniel Park

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## EDUCATION

### Western University

Sept. 2021 – Apr. 2026

*Bachelor of Science, specialization in **Computer Science***

- **Relevant Coursework:** Machine Learning, Databases, Operating Systems, Computer Networks, Software Engineering

## EXPERIENCES

### Machine Learning Intern

Sept. 2025 – Apr. 2026

*HEAL (Python, AWS S3, Airflow, Google Places API, Scikit-Learn, XGBoost)*

- Architected a **cloud-native ETL pipeline** on **AWS S3** via **Apache Airflow** and the **Google Places API** to detect **spatial anomalies**, processing **25K+** retail records across **80+** Canadian cities.
- Engineered **Python** workflows to sanitize **unstructured text** and normalize **metadata**, reducing **dataset anomalies** and inconsistencies by **30%**.
- Implemented **MLOps practices**, **versioned storage**, and rigorous **schema validation** to establish **data governance** and ensure pipeline reliability.
- Trained **predictive Scikit-Learn** and **XGBoost** models (**clustering, time-series forecasting**) to analyze **retail density trends**, deploying **visualizations** for **data-driven** strategic decisions.

### Data Science Intern

Sept. 2024 – Aug. 2025

*interRAI (Python, PySpark, Spark SQL)*

- Managed **distributed PySpark ETL** workflows to validate and transform **250K+** clinical assessment records, ensuring **high-fidelity inputs** for health research.
- Modernized legacy **data pipelines** into **automated Spark SQL workloads**, reducing execution runtime by **30%** and lowering manual processing overhead.
- Documented core **Spark architectures** and led technical training, accelerating onboarding and streamlining **cross-functional collaboration**.

### Software Developer Intern

May 2024 – Aug. 2024

*BERL (React, FastAPI, CSS Flexbox/Grid, Cascade CMS)*

- Refactored **15+** legacy **React components** and **Cascade CMS templates** via **CSS Flexbox/Grid**, reducing **system latency** across **25+** production pages.
- Built **6+** **FastAPI endpoints** for **administrative workflows**, enforcing **accessibility standards** and optimizing **backend form handling**.

### Software Developer Intern

Sept. 2023 – Apr. 2024

*Cultureplex (Python, SQL, PostgreSQL, LLMs, Matplotlib, Seaborn)*

- Engineered **Python** and **SQL ETL pipelines**, leveraging **LLMs (Generative AI)** to transform **1K+** unstructured research records into **searchable metadata** and **structured PostgreSQL tables**.
- Optimized **PostgreSQL databases** via advanced **data modeling, indexing, and query tuning**, improving **retrieval latency** by **35%**.
- Generated **Matplotlib** and **Seaborn data visualizations** to translate **aggregated data** for **cross-functional teams**, accelerating academic publications.

## PROJECTS

### Canadian Ski Resorts Powder Predictor | *Airflow, PySpark, AWS S3/RDS, XGBoost, Flask, React*

- Architected an end-to-end **Airflow, AWS S3, and PySpark ETL pipeline** to process **62,000+** historical records and automate daily ingestion of **7-day weather forecasts** for **18 Canadian ski resorts** via **Open-Meteo REST APIs**.
- Engineered temporal features, including **3-day rolling snowfall** and **freeze-thaw cycles**, to train an **XGBoost Regressor** predicting a "Powder Score" (**1-10**) with an **MAE of 0.01**, storing inferences in an **AWS RDS PostgreSQL** warehouse.
- Built a **Flask REST API** and a dark-mode **React** dashboard to expose powder scores, resort comparisons, and weather-driven recommendations across **15+** resort views.

## TECHNICAL SKILLS

**Languages:** Python, R, SQL, Java, C/C++, JavaScript, Bash, PHP, HTML/CSS

**ML & Data:** PySpark, Spark SQL, Airflow, pandas, NumPy, Polars, SciPy, Scikit-Learn, XGBoost, PyTorch, TensorFlow, LLMs / Generative AI, Matplotlib, Seaborn, SHAP, UMAP, Yellowbrick, Jupyter, ETL/ELT

**Backend & Cloud:** AWS S3/RDS, Azure, GCP, Flask, FastAPI, Django, .NET, React, PostgreSQL, MySQL, MSSQL, SQLite, MongoDB, DynamoDB, Snowflake, Elasticsearch, Lucene, MapReduce, Docker, Git, Maven, Regex, Claude Code

**Certifications:** AWS Machine Learning Engineer Certificate, Meta Back-End Developer Professional Certificate